

PS06 : Negotiating AI Memory

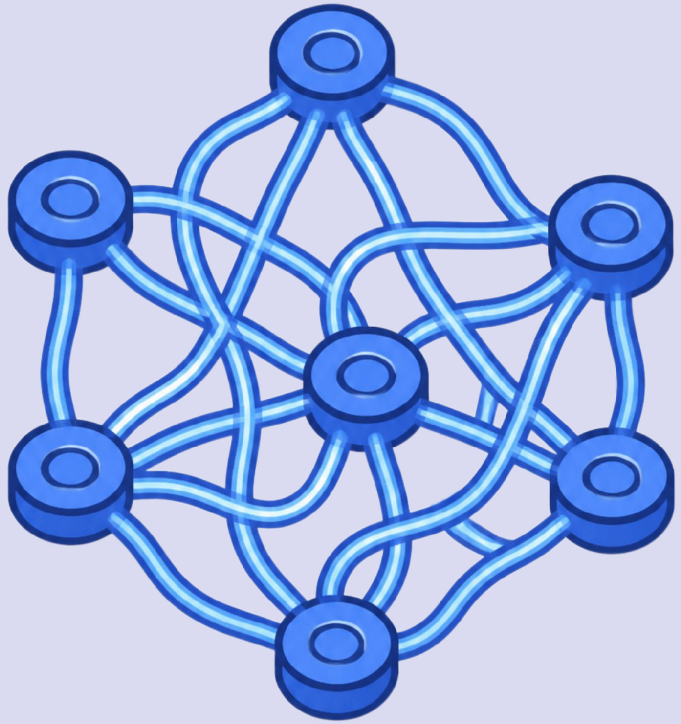
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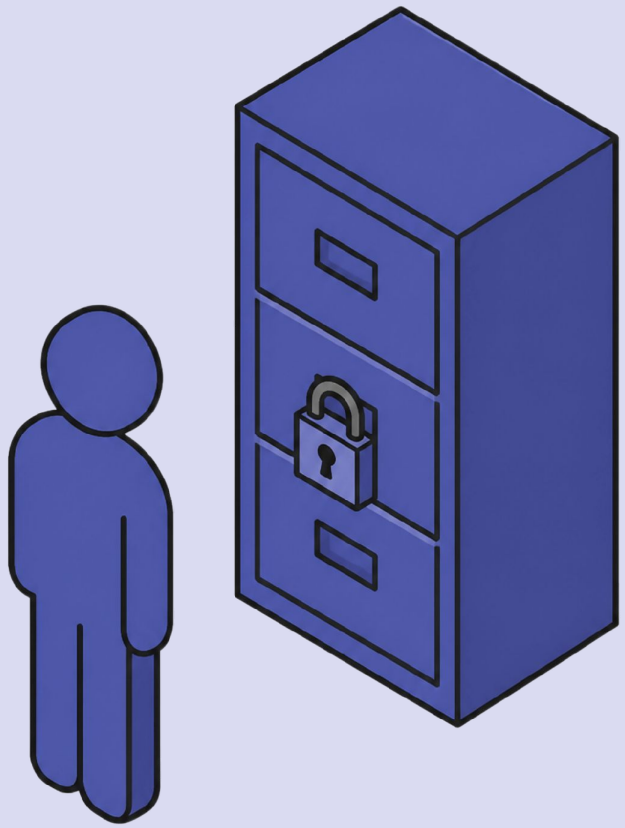
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The Problem: Silent, Uncontrolled Memory

AI assistants accumulate memory silently — users cannot see, correct, or control what's remembered. Three critical harms emerge from this invisible process:

Lack of Transparency

Unpredictable behavior caused by hidden, unreviewed memories

Temporal Validity Problem

Outdated facts resurface at the wrong moment

Sensitive Information Exposure

Private disclosures bleed into unintended conversations



REAL INCIDENT

A Real Incident

A team member's research paper was marked **"complete"** in AI memory — while still actively in progress.

That stale fact propagated into an AI-generated CV and surfaced during a live interview.

This is a **temporal-validity failure**, not an accuracy failure. The system needs to know *when* facts are true — not just whether they were ever true.

Mapping User Needs to Implemented Solutions

USER NEED	IMPLEMENTED SOLUTION
Transparency of Memory	Shows what AI remembers and how memories are used.
Easy Access & Organization	Memory panel with **List and Graph views**.
Transparency in Data Usage	Shows **where memories came from** and which memories influenced responses.
User Control over Memory	Users can **add, edit, reject, or rescope** memories.
Privacy Protection	**PII detection**, redaction warnings, and **session-only memory**.
Bounded AI Relationship	Separates **stated vs. inferred memories** and verifies memories for important outputs like CVs.

Research Insights: Key Findings



Need for Transparency

Users want to understand how their data is processed and stored — not discover it after harm occurs



Desire for User Agency

"I should be able to decide the level of detail" — users demand granular, per-memory control



Privacy Concerns

Users fear data leakage across contexts and unintended memory persistence across sessions



Accessibility Matters

Memory features must be reachable without a mouse and without sight — for every user

Design Implications

Our research led to three non-negotiable design principles that guide every UI decision in AI Memory Manager:



1 Visible at the Moment of Formation

Memory formation must be surfaced *when it happens* — not auditable only in retrospect

2 Control Without Fatigue

Most items pass silently; user attention is spent only where it genuinely matters

3 Full Provenance

Every memory must be traceable back to the exact message that created it

Our Solution: Negotiated Memory

Instead of silent accumulation, AI Memory Manager introduces a **real-time negotiation loop** between the user and the system.

1

Inline Memory Cards

Appear as the system learns — accept, reject, or edit in one tap

2

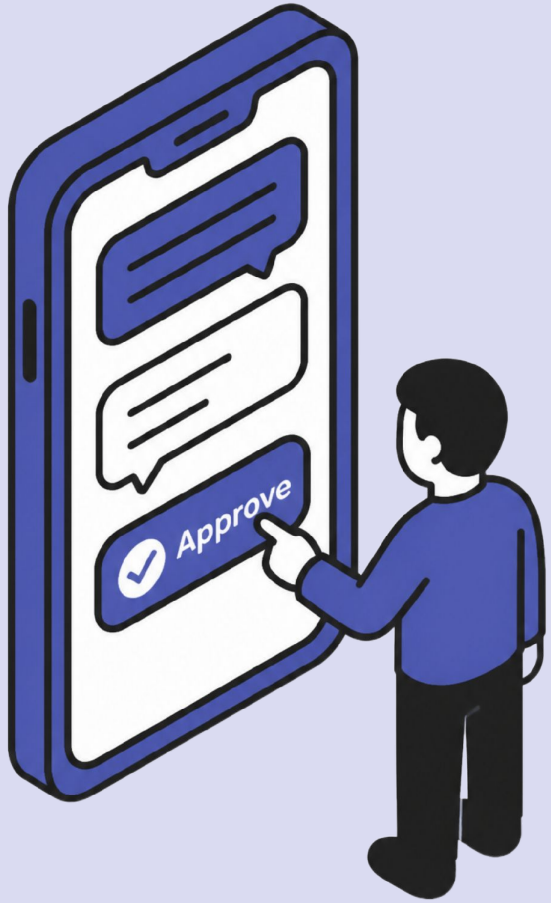
Session-Only Mode

Mark memories as temporary to prevent any persistence beyond the conversation

3

Response Attribution

See which memories shaped each response — and revoke them instantly



Key Features Shipped

1 Real-Time Negotiation Loop

Inline decision-making at the moment of memory formation — no settings diving required

2 Assertion Status

Tag memories as *in-progress*, *completed*, *planned*, or *abandoned* for full temporal validity

3 Provenance-Aware Deletion

Preview what else is affected before confirming any delete — no surprises

4 Stake-Proportional Verification

High-stakes artifacts like CVs and credentials trigger elevated verification before use

Why This Matters

AI memory is becoming the substrate of digital identity. Getting it wrong has real consequences. Getting it right builds the trust that makes AI genuinely useful.

Users Back in Control

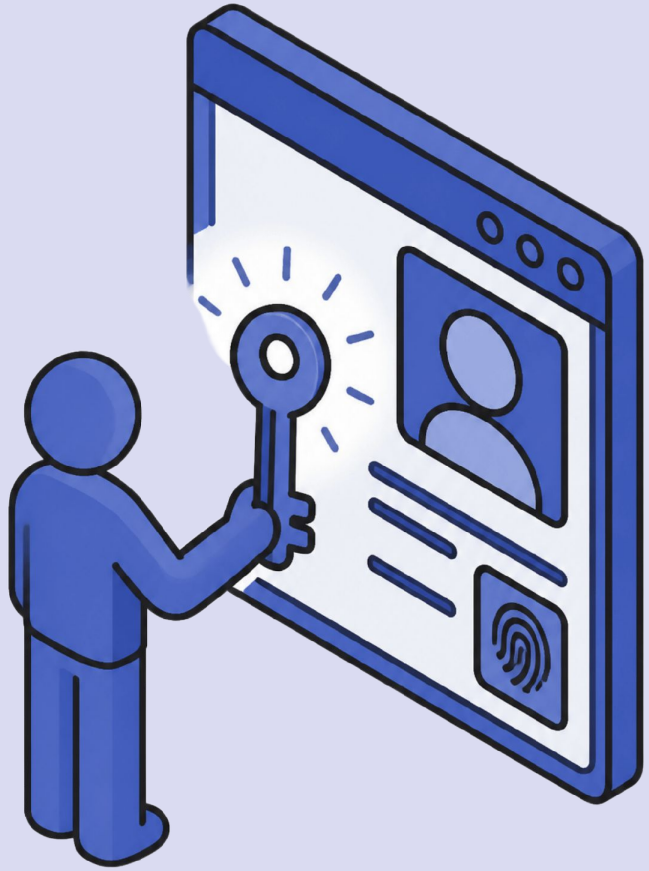
Puts people in charge of their own digital identity — not the AI's hidden assumptions

Solves the CV Problem

Memory stays accurate, time-stamped, and contextually appropriate — no more stale facts at interviews

Trust Through Transparency

Builds genuine trust in AI systems by making every memory visible, traceable, and revocable



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